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Chapter 3

Save To Media



Introduction To Save To Media

You can perform three kinds of library save to media. The method used is determined by Housekeeping's Control Information (see Control Information section in the Control Details topic). The methods are:

1. Select - each library is saved to a separate series of tapes, cartridges or diskettes. The operator is prompted to save libraries individually. Once the media volume ID has been assigned, all subsequent saves of the library will prompt the operator to load that volume again. Therefore the same media volume is used.
2. Mass - Housekeeping saves as many libraries to a media volume as it can. This is calculated using library sizes and the media size and density. A library is not split over two volumes. Libraries are thus organised into groups. The operator is prompted to save these groups of libraries. A media volume ID must be entered for each save. This ID is unique and for a volume to be re-used it must be freed.
3. Cycle - specific media volumes are used on specific days to perform the mass save, the number of days in the cycle is pre-set in Housekeeping's Control Information. For example, you can use a five day rotation cycle. This might correspond to delay saving from Monday (Day 1) to Friday (Day 5). This uses five sets of off-line storage media, one set for each day. The operator is prompted to load the same set of volumes in the same order as on this day in the previous cycle. The media are freed up automatically.



Note: As the save media is freed automatically Housekeeping does not check if the media contains the latest saves of any libraries. Therefore you should set the libraries to Force Save=1 and Save Frequency equal to the number of days in the cycle. Otherwise you could lose your last save of a library. See Control Information section in the Control Details topic and the Libraries section of the Definitions topic.



Tip: Folders and off-site libraries can only be saved to media by a mass save routine.



Note: For easy identification it is recommended that the volume ID's for library, folder and off-site saves are distinguished by a separate ID prefix.

Libraries



Select option 1 from Save To Media and press **Enter**.

The screen displayed is determined by the Normal Backup Method selected in Housekeeping's Control Information (see Control Information section in the Control Details topic).

- 1 - If you perform Select backups, refer to Library Backup - Select Method.
- 2 - If you perform Mass backups, refer to Library Backups - Mass Method.
- 3 - If you perform Cyclic backups, refer to Library Backups - Cyclic Method.

Library Backup - Select Method



Select option 1 from Save to Media and press **Enter**.

HK800	House keeping	System: STUDB30A
Libraries Backups Routine Entry - Select Method		
You have selected the option to run the Libraries Backups routine. The last Library Auto-backup routine was started on 24/06/91 at 20:27:02 ran successfully and finished on 24/06/91 at 20:38:21		
Any saves to Save Files will have been made to Library OSLSAVF The Auto-backup routine will, if possible, have saved all Libraries to Save Files. The Backup List will show whether or not this has happened.		
By taking option 1 below you will be presented with a list of the Libraries which were not saved to Save Files by the Auto-backup routine. If there are none then Backups may be made from the Save Files by taking option 2 below.		
Type option, press ENTER		
1=Save Libraries Not Saved to Save Files. 2=Save Libraries from Save Files. Option _		
F3=Exit		

Fields:

Option

Enter 1 to save those libraries to media which have not been saved to save files by Housekeeping's overnight routines (see Libraries section in the Execute Day-End Functions topic).

Enter 2 to save those libraries to media which have been saved to save files by Housekeeping's overnight routines (see Libraries section in the Execute Day-End Functions topic).

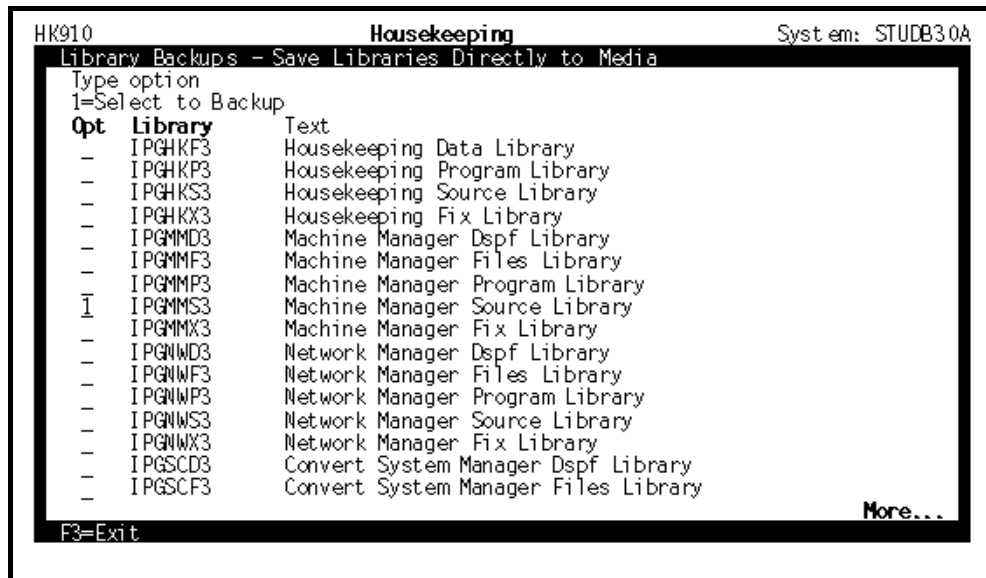


Enter the required option and press **Enter**.

Select Library



Enter the required option and press **Enter**.



Enter 1 against the library which you wish to save to media.



Tip: If there are no libraries to save then a screen will be displayed informing you of this.

Select Media



Select the library.

If the library has been previously saved by Housekeeping, you must load the specific media volume used.

If the library has not been previously saved by Housekeeping, select the media type required and the volume ID. You may optionally specify its location.

```
HK010                               Housekeeping                               System: STUDB30A
Library Backups - Save Libraries Directly to Media

You have selected to save the following:-

Library  IPGMS3      Machine Manager Source Library

No details are available to specify what Media Type, Volume ID, etc.,
you need to save the above Library.

If you wish to save this Library specify the relevant details below.

Type in details, and press ENTER to accept

Media type  _      1=Cartridge, 2=Diskette, 3=Tape
Volume ID   _____
Location    . _____

F3=Exit    F12=Previous
```

Fields:

Media Type

Enter 1 for cartridge, 2 for diskette or 3 for tape.

Volume Id

Enter the volume ID of the media to be used. This must be a unique media ID.

Location

Enter the location of the media. This is your identifier for the location of the volume, for example, shelf no, position.



Tip: You can initialise the media to the specified volume ID before performing the save by selecting **F24**.



Load the required media and press **Enter** to perform the save.

Library Backups - Mass Method



Select option 1 from Save to Media and press **Enter**.

A list of the library save groups is displayed. Housekeeping organises libraries into these save groups so that a group of libraries can be saved onto a single media volume. A library will not be saved across two volumes.

HK955

Housekeeping

System: STUDB3.0A

Manage Library Saves - Mass Backups

You have selected the option to perform the MASS Backup to CART RIDGE.
You require 1 Cartridges to perform the outstanding backups.

Type option

1=Select for Backup

Opt	Save Group	Size (Mb)	No.of Libraries
1	2	4	1

F3=Exit

Bottom

Fields:

Opt

Enter 1 against the save group you want to save to media.



Tip: If there are no libraries to save a screen is displayed informing you of this.

Group Save Details/Select Media

This displays details of the save group you selected.

```
HK055          Housekeeping          System: STUDB30A
Manage Library Saves - Mass Backups

You have selected to save group      2

This save group has a total size of    4 Mb
                        and contains    1 Libraries.

Enter below the Volume ID and Location of the save CARTRIDGE.

Type in details, and press ENTER to perform save

Volume ID  _____
Location   _____

F3=Exit   F12=Previous   F24=Initialise Media
```

Fields:

Volume Id

Enter the volume ID of the media to use. This must be a unique media ID.



*Note: If you wish to re-use existing media see-
Free Library Media. section .*

Location

Enter the location of the media. This is your identifier for the location of the volume for example, shelf number, position.

Functions:

F24

Select **F24** to initialise the media before performing the save.



Load the required media and press **Enter** to perform the save.

After the save has been completed you will return to the save group selection screen unless there are no more save groups to save.



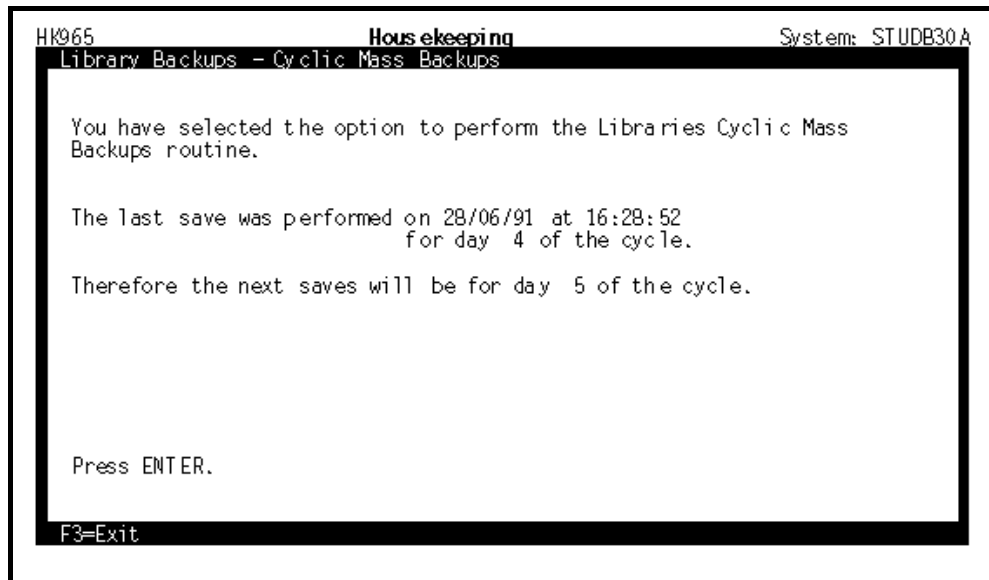
Note: An audit report is produced automatically listing the libraries saved to each media volume and the sequence in which they were saved.

Library Backups Cyclic Method



Select option 1 from Save to Media and press **Enter**.

This screen displays details of the last save in the cycle. A new day in the cycle is assumed if the last save was carried out prior to today's date.



Press **Enter** to continue with the save.

Library Backups Cyclic Details Screen

A list of the library save groups is displayed. Housekeeping organises libraries into these save groups so that a group of libraries can be saved onto a single media volume. A library will not be saved across two volumes.

```
HK965                Housekeeping                System: STUDB30 A
Library Backups - Cyclic Mass Backups

You have selected the option to perform the Libraries Cyclic Mass
Backups routine to CARTRIDGE.
You require      1 Cartridges to perform the outstanding backups.

Press ENTER to Save the Group highlighted
Opt  Save Group  Size (Mb)  No. of Libraries  Volume ID  Location
  1      2         4           1                1
                                     Bottom

F3=Exit
```

Fields:

Opt

Enter a 1 against the save group you wish to save to media.



Tip: If there are no libraries to save a screen will be displayed informing you of this.

Group Save Details/Select Media



The Library Backups - Cyclic Mass Backups window is displayed.

This displays details of the save group you selected.

HK965 Housekeeping System: STUDB30.A
Library Backups - Cyclic Mass Backups

You have selected to save group 2 which is volume 1 of day 5
This save group has a total size of 4 Mb
and contains 1 Libraries.

Enter below the Volume ID and location of the save CARTRIDGE if not
previously selected. Remember that this will FREE THE MEDIA specified.

Type in details, and press ENTER to perform save

Volume Id ____
Location ____

F3=Exit F12=Previous F24=Initialise Media

Fields:

Volume Id

Enter the volume ID of the media to be used. This must be a unique media ID.



*Note: To re-use existing media see section -
Free Library Media.*

Location

Enter the location of the media. This is your identifier for the location of the volume, for example, shelf number or position.

Functions:

F24

Press **F24** if you wish to initialise the media before performing the save.



Load the required media and press **Enter** to perform the save.

After the save has been completed you will return to the save group selection screen unless there are no more save groups to save.



*Note: An audit report is produced automatically
listing the libraries saved to each media volume
and the sequence in which they were saved.*

Archiving Libraries

Once a library has been created by the Housekeeping module it can remain on the IBM AS/400 indefinitely. However, you may want to remove the library from the machine. Housekeeping provides two methods for removing libraries from the system in a controlled and fully audited manner. They can be archived or deleted, see the Library Deletion section in the Definitions topic.

If you want to remove a library which is likely to be used in the future, then it should be archived. Two saves to off-line storage are made. This reduces the dependence on a single storage media in the unlikely event that it is faulty. A library can be flagged for archiving via the Display User Libraries facility (see User Libraries section in the Operational Enquiries topic).

All libraries flagged for archiving will be saved as save files by the following Housekeeping overnight routine (see Libraries section in the Execute Day-End Functions topic).



To save archived libraries from save files to media select option 2 from Save to Media and press **Enter**.

The next screen displayed is determined by the Normal Backup Method selected in Housekeeping's Control Information (see Control Information section in the Control Details topic).

- 1 - If you perform Select backups then refer to Library Backup - Select Method.
- 2 - If you perform Mass backups then refer to Library Backups - Mass Method.



Note: If you have Housekeeping Control Information to perform cyclic backups then archived libraries are saved using the select method.

Archive Library Backup Select Method



Select option 2 from Save to Media and press **Enter**.

```
HK700                               House keeping                               System: STUDB30A
Libraries Archiving Routine Entry - Select Method

You have selected the option to run the Libraries Archiving routine.
There are outstanding Libraries to be Archived.

Any Saves to Save Files will have been made to Library OSLARCLIB
The Auto-backup routine will, if possible, have saved all Libraries to
Save Files. The Archive Backup List will show whether or not this has
happened.

By taking option 1 below you will be presented with a list of the
Libraries which were not saved to Save Files by the Auto-backup routine.
If there are none then Backups may be made from the Save Files by taking
option 2 below.

Type option, press ENTER

1=Save Libraries Not Saved to Save Files.
2=Save Libraries from Save Files.
Option _

F3=Exit
```

Fields:

Option

Enter 1 to save those libraries to media which have not been saved to save files by Housekeeping's overnight routines (see Libraries section in the Execute Day-End Functions topic).

Enter 2 to save those libraries to media which have been saved to save files by Housekeeping's overnight routines (see Libraries section in the Execute Day-End Functions topic).



Enter the required option and press **Enter**.

Select Library



Enter the required option and press **Enter**.

Opt	Library	Saves	Text
I	IPGAMD3		Application Manager Display
I	IPGAMF3		Application Manager Files
-	IPGAMP3		Application Manager Programs



Enter 1 against the library which you wish to save to media.



Tip: If there are no libraries to save then a screen will be displayed informing you of this.

Select Media

 Select the library.



The screenshot shows a terminal window titled "Housekeeping" with the system identifier "ST UDB30A". The main heading is "Library Archiving - Archive Save File to Media". The text inside the window reads: "You have selected to save the following:-", "Library **IPGAMF3** Application Manager Files", and "If you wish to save this library specify the relevant details below." Below this is a prompt: "Type in details, and press ENTER to accept". There are three input fields: "Media Type" with a dropdown menu showing "1=Cartridge, 2=Diskette, 3=Tape", "Volume ID" with a blank line, and "Location" with a blank line. At the bottom, it says "F3=Exit F12=Previous".

Fields:

Media Type

Enter 1 for cartridge, 2 for diskette or 3 for tape.

Volume Id

Enter the volume ID of the media to be used. This must be a unique media ID.

Location

Enter the location of the media. This is your identifier for the location of the volume, for example, shelf number, or position.

 Tip: You can initialise the media to the specified volume ID before performing the save by selecting **F24**.

 Load the required media and press **Enter** to perform the save.

Archive Library Backups Mass Method



Select option 1 from Save to Media and press **Enter**.

A list of the library save groups is displayed. Housekeeping organises libraries into these save groups so that a group of libraries can be saved onto a single media volume. A library will not be saved across two volumes.

HK715 Housekeeping System: STUDB3.0A

Libraries Archiving - Mass Archives

You have selected the option to perform the MASS Archive to CARTRIDGE.
You require 6 Cartridges to perform the outstanding archiving.
REMEMBER - EACH SAVE GROUP MUST BE SAVED TWICE

Type option

1=Select for Archiving

Opt	Save Group	Size (Mb)	No. of Libraries	No. of Saves
1	1	2	2	0
2	2	123	1	0
3	3	7895	1	0

F3=Exit Bottom

Fields:

Opt

Enter a 1 against the save group you wish to save to media.



Tip: If there are no libraries to save a screen is displayed informing you of this.

Group Save Details/Select Media

This displays details of the save group you selected.

Fields:

Volume Id

Enter the volume ID of the media to be used. This must be a unique media ID



Note: If you wish to re-use existing media see - Free Library Media section.

Location

Enter the location of the media. This is your identifier for the location of the volume.

Functions:

F24

Press **F24** to initialise the media before performing the save.



Load the required media and press **Enter** to perform the save.

After the save has been completed you will return to the save group selection screen unless there are no more save groups to save.



Note: An audit report is produced automatically listing the libraries saved to each media volume and the sequence in which they were saved.

Folders



Select option 3 from Save to Media and press **Enter**.

The Housekeeping overnight routines will have calculated which folders are due to be saved see Folder section in the Execute Day-End Functions topic.

A list of the folder save groups is displayed. Housekeeping organises folders into these save groups so that a group of folders can be saved onto a single media volume. A folder will not be saved across two volumes.

HK755

Housekeeping

System: STUDB3.0A

Folder Backups - Mass Backups

You have selected the option to perform the MASS Backup to CART RIDGE.
You require 1 Cartridges to perform the outstanding backups.

Type option

1=Select for Backup

Opt	Save Group	Size (Mb)	No.of Folders
1	1	8	3

Bottom

F3=Exit

Fields:

Opt

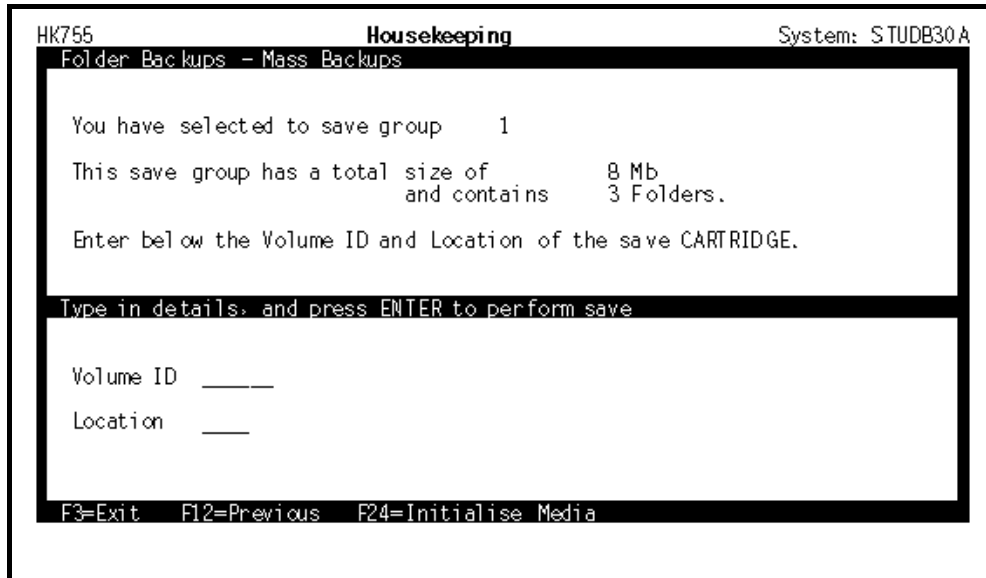
Enter 1 against the save group you want to save to media.



Tip: If there are no folders to save a screen is displayed informing you of this.

Group Save Details/Select Media

This displays details of the save group and you have selected.



The screenshot shows a terminal window titled 'Housekeeping' with 'HK755' on the left and 'System: STUDB30 A' on the right. The main menu is 'Folder Backups - Mass Backups'. The text inside the window reads: 'You have selected to save group 1', 'This save group has a total size of 8 Mb and contains 3 Folders.', and 'Enter below the Volume ID and Location of the save CARTRIDGE.'. Below this is a prompt 'Type in details, and press ENTER to perform save'. There are two input fields: 'Volume ID' followed by a line of underscores, and 'Location' followed by a line of underscores. At the bottom, a status bar shows 'F3=Exit F12=Previous F24=Initialise Media'.

Fields:

Volume Id

Enter the volume ID of the media to be used. This must be a unique media ID



Note: If you wish to re-use existing media see - Free Folder Media section.

Location

Enter the location of the media. This is your identifier for the location of the volume.

Functions:

F24

Press **F24** to initialise the media before performing the save.



Load the required media and press **Enter** to perform the save.

After the save has been completed you will return to the save group selection screen unless there are no more save groups to save.



Note: An audit report is produced automatically listing the folders saved to each media volume and the sequence in which they were saved.

Off Site Libraries



Select option 4 from Save to Media and press **Enter**.

The Housekeeping overnight routines will have saved a group of libraries belonging to an off-site save group to a save file. The off-site save group which will have been saved is determined by the Select Off-Site Save Group function (see Select Off-Site Group section in the Execute Day-End Functions topic) which must be run before the initiation of the Off-Site Libraries Day-End routine (see Off-Site Libraries section in the Execute Day-End Functions topic).

A list of the archived library save groups is displayed. Housekeeping organises archived libraries into these save groups so that a group of archived libraries can be saved onto a single media volume. An archived library will not be saved across two volumes.

```
HK765                               Housekeeping                               System: STUDB3.0A
Library Save Groups Backups - Mass Backups

You have selected the option to perform the MASS Backup to CART RIDGE.
You require      1 Cartridges to perform the outstanding backups.

Type option
1=Select for Backup
Opt  Save Group  Size (Mb)  No.of Libraries
  1      1         1         2

F3=Exit                                Bottom
```

Fields:

Opt

Enter 1 against the save group you want to save to media.



Tip: If there are no archived libraries to save a screen will be displayed informing you of this.

Group Save Details / Select Media

This displays details of the save group and you have selected.

The screenshot shows a terminal window titled 'Housekeeping' with 'System: STUDB30 A' in the top right. The main title is 'Library Save Groups Backups - Mass Backups'. The text inside the window reads: 'You have selected to save group 1', 'This save group has a total size of 1 Mb and contains 2 Libraries.', and 'Enter below the Volume ID and Location of the save CARTRIDGE.' Below this is a prompt 'Type in details, and press ENTER to perform save'. There are two input fields: 'Volume ID' followed by a line of underscores, and 'Location' followed by a line of underscores. At the bottom, there are function key shortcuts: 'F3=Exit', 'F12=Previous', and 'F24=Initialise Media'.

Fields:

Volume Id

Enter the volume ID of the media to be used. This must be a unique media ID.



Note: If you wish to re-use existing media see - Free Library Media section.

Location

Enter the location of the media. This is your identifier for the location of the volume.

Functions:

F24

Press **F24** if you wish to initialise the media before performing the save.



Load the required media and press **Enter** to perform the save.

After completing the save you will return to the Save Group Selection screen, unless there are no more groups to save.



Note: An audit report is produced automatically listing the archived libraries saved to each media volume and the sequence in which they were saved.

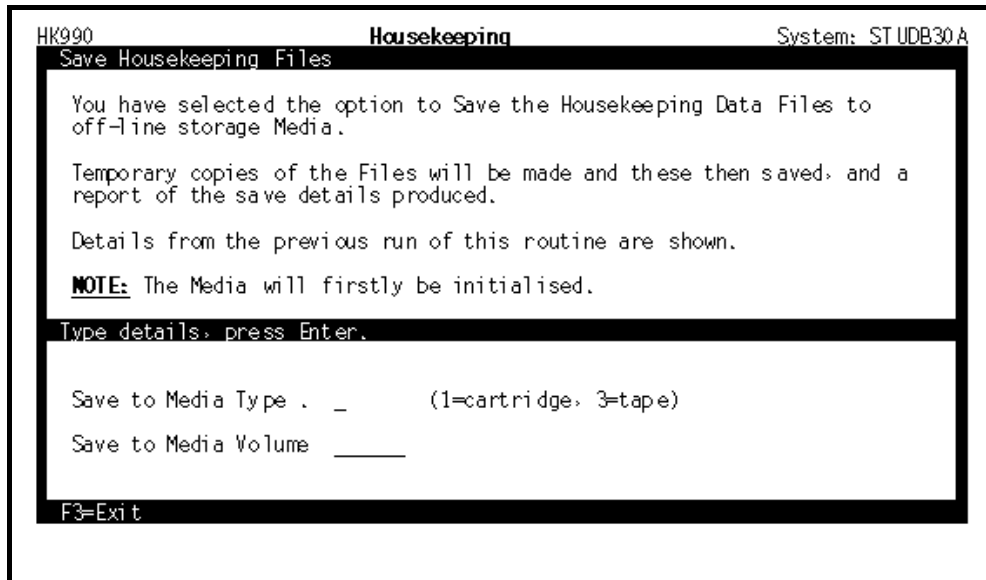
Housekeeping Files

One of the principal functions of Housekeeping is to produce an accurate history of library saves. This log is important if you need to restore a library. It is of even greater importance if you need to restore a system. In such a scenario Housekeeping needs to be restored initially to tell the Installation Manager where to find the latest save of each library, so that they can be restored. The most accurate picture of the saves performed is immediately after all the backups have been completed. If you rely on the save of the Housekeeping files library by the Housekeeping overnight routine, details of the day's backups are not reflected in the current save of the Housekeeping files.

Therefore it is recommended that the Housekeeping file save routine is performed after the backups have been completed as a separate activity. The option to restore Housekeeping files, as one of the first steps in the recovery process, is also available (see Housekeeping Files section in the Restore From Media topic).

Save Housekeeping Files Screen

-  Select option 5 from Save to Media and press **Enter**.



```
HK990                Housekeeping                System: STUDB30 A
Save Housekeeping Files

You have selected the option to Save the Housekeeping Data Files to
off-line storage Media.

Temporary copies of the Files will be made and these then saved, and a
report of the save details produced.

Details from the previous run of this routine are shown.

NOTE: The Media will firstly be initialised.

Type details, press Enter.

Save to Media Type . _      (1=cartridge, 3=tape)
Save to Media Volume  ____

F3=Exit
```

Fields:

Save To Media Type

Enter 1 to save to cartridge or 3 to save to tape.

Save To Media Volume

Enter the volume ID of the media to be used. This must be a unique media ID



Note: If the media volume you enter is already in use then you cannot use it to save Housekeeping files. If the media has not been used before or has been used previously for Housekeeping files it is now initialised.



Press **Enter** to perform the save.

Free Library Media



Select option 6 from Save to Media and press **Enter**.

Use this function to free up media which you have previously saved libraries onto. It is possible only to free up media which contains no active libraries, that is all the libraries must have had subsequent saves from save files to media performed.

A list of all the media available to free is displayed.

HK505

Housekeeping

Syst em: STUDB3.0A

Select Library Save Media to Free

Type options, and press ENTER

1=Select to free

Opt	Media Type	Volume ID	Location	Backup Type
-	CART RID GE	MB0003	T525	MASS
-	CART RID GE	MB0004	S857	MASS
-	CART RID GE	MB0008	R906	MASS
-	CART RID GE	MB0012	T258	MASS
-	CART RID GE	MB0013	Y826	MASS
-	CART RID GE	MB0027	S565	MASS
-	CART RID GE	MB0850	D552	MASS

F3=Exit F15=Media to Keep

Bottom

Fields:

Opt

Enter 1 against the media you want to free up for future use.



A prompt screen is displayed for you to confirm the selections you have made.

Reports

Reports detailing the following are automatically produced.

- Media available to free.
- Media which contains active libraries, and so must be kept.
- Media you have selected to be freed.

Functions:

F15

Use this function to display the Display Library Save Media Screen.

Display Library Save Media Screen

This screen details the media to keep, that is media which contains at least one active library.

HK505	Housekeeping				System: STUDB30A
Display Library Save Media to Keep					
Type options, and press ENTER					
1=Display details					
Opt	Media Type	Volume ID	Location	Backup Type	No. of active Libraries
I	CARTRIDGE	MB0500	ST48	MASS	1
I	CARTRIDGE	MB0501	ST48	MASS	1
-	CARTRIDGE	MB0503	ST52	MASS	30
					Bottom.
F3=Exit F12=Previous F15=Media to Free					

Fields:

Opt

Enter alongside a media volume to display full information about the libraries it contains.



Select a media volume.

Display Media Volume Details Screen

This screen displays the details for the selected media volume. This includes all of the libraries contained on the media.

The screenshot shows a terminal window titled "Housekeeping" with the system identifier "STUDB30A". The main heading is "Display Media Volume Details". Below this, the following information is displayed: Media Type **CARTRIDGE**, Volume ID **MB0501**, Location **ST4 B**, and Backup Type **MASS**. A prompt "Type options, and press ENTER" is shown. Below the prompt, the option "1=Single Force Save" is listed. A table of libraries is displayed with columns "Library", "Text", and "Active Messages". The table contains one entry: "0 Library" with "Text" "Application Manager Progr *YES" and "Active Messages" "F S". At the bottom left, it says "F12=Previous" and at the bottom right, it says "Bottom".

```
HK505                               Housekeeping                               System: STUDB30A
Display Media Volume Details
Media Type  CARTRIDGE  Volume ID  MB0501  Location  ST4 B  Backup Type  MASS
Type options, and press ENTER
1=Single Force Save
0 Library   Text               Active Messages  F
_ IPGAMP3   Application Manager Progr *YES                      S
F12=Previous                                     Bottom
```



If you need media to free up for subsequent saves, select an active library to force save by entering 1 alongside the library.



Note: The force save will save the library(ies) selected as part of the next Housekeeping overnight routines (see Libraries section in the Execute Day-End Functions topic). Once those saves have been saved to media then the library is no longer active on this particular media volume, and this can be freed.

Free Folder Media



Select option 7 from Save to Media and press **Enter**.

Use this to free up media which you have previously saved folders onto. You can also free up media which contains no active folders, that is all the folders must have had subsequent saves from save files to media performed.

A screen containing a list of all media available to free will be displayed:

HK515
Housekeeping
System: STUDB30A

Select Folder Save Media to Free

Type options, and press ENTER

1=Select to free

Opt	Media Type	Volume ID	Location
-	TAPE	FB0H03	FT25

F3=Exit
Bottom

Fields:

Opt

Enter 1 against the media you want to free up for future use.



A prompt screen will then be displayed for you to confirm the selections you have made.

Reports

Reports detailing the following are automatically produced:

- Media available to free.
- Media which contains active folders and so must be kept.
- Media you have selected to be freed.